

# CMSE 801

Introduction to Computational Modeling & Data Analysis

## **Welcome!**

**Please double-check your email for  
information about this course**

Devin W. Silvia, PhD  
Director of Undergraduate Studies  
Dept. of Computational Mathematics, Science and Engineering

# THE **THREE THREADS** OF CMSE 801

- **Coding in Python:** By the end of the CMSE 801 you will be a proficient programmer/coder in Python. You should be able to use Python to help you in all of future classes.
- **Modeling:** You will learn how to build models that allow you to quantitatively solve complex problems in science, engineering and beyond. Combined with the coding aspect of the course, these models will be turned into computational models.
- **Data Analysis:** You will learn how to handle real-world data: find it, analyze it, and draw quantitative conclusions from it. You will use your coding skills for this analysis and build models that reveal deeper insights into the data.



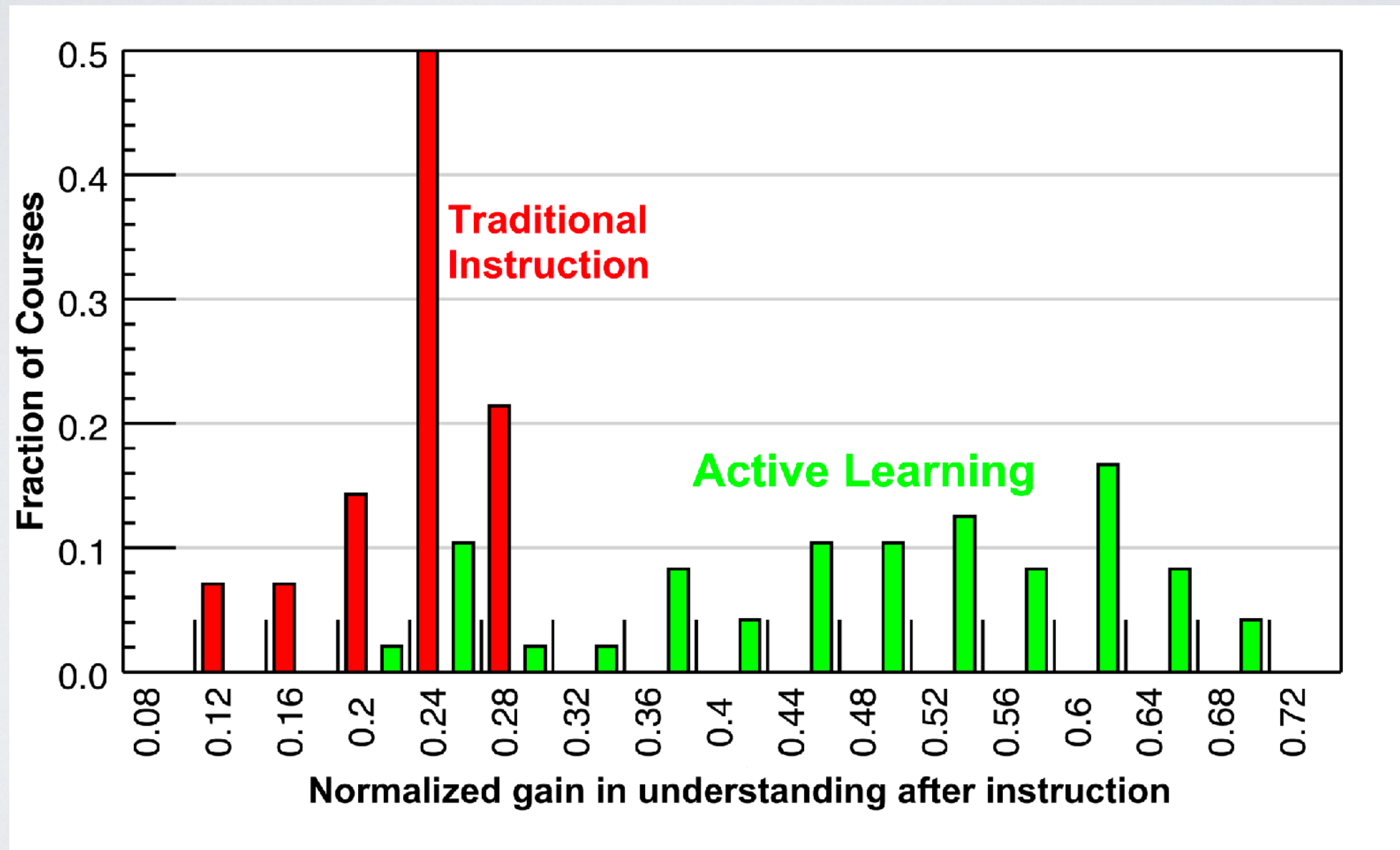
# COURSE STRUCTURE

- **Flipped classroom:** The class is structured to have pre-class assignments, which you will do before coming to class and in-class assignments done during class, often as part of a group.
- There is homework every 2-3 weeks.
- Also, the only way to learn to code is to code. We will code in class every day. Bring your computer with you every day; we will have a few backups if you need one.
- There will be a handful of quizzes throughout the semester.
- Projects will be presented at the end of the semester, although we will start planning your projects earlier in the semester.

# A FEW NOTES ABOUT THE FLIPPED CLASSROOM

- The flipped classroom environment will like feel very different than what you may be used to.
- Learning takes place in a collaborative manner in which you learn by **doing**, using knowledge you acquire before coming to class
- Pre-class assignments are critical for maximizing your in-class experience

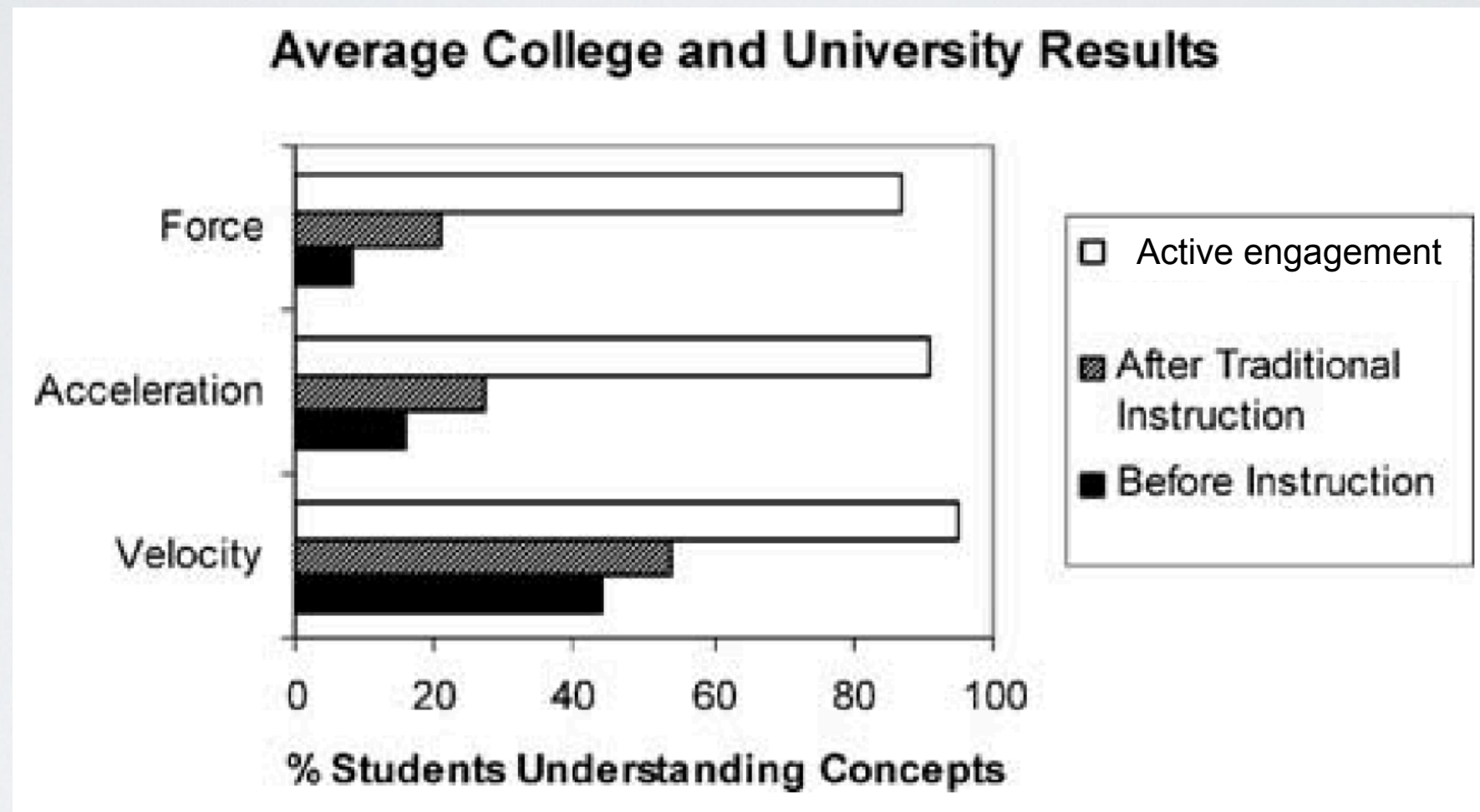
# INTERACTIVE ENGAGEMENT LEADS TO MORE LEARNING!



Hake, "Interactive-engagement versus traditional methods: A six-thousand-student survey ...." American Journal of Physics. 1998.



# INTERACTIVE ENGAGEMENT LEADS TO MORE LEARNING!



Active-engagement vs. traditional instruction for improving students' conceptual understanding of basic physics concepts (taken from Laws et al., 1999)

# GETTING THE MOST OUT OF THE FLIPPED CLASSROOM

- You are responsible for monitoring your own learning — only **you** can ensure that you are mastering the content!
- On D2L, in the “Course Information” section, there is a PDF called “classroom\_participation”. Take a moment to review this document

# GRADES

|            | pre-class assignments | participation, attendance, in-class assignments | quizzes | homework | final project |
|------------|-----------------------|-------------------------------------------------|---------|----------|---------------|
| percentage | 15%                   | 15%                                             | 20%     | 25%      | 25%           |

|     |      |
|-----|------|
| 4.0 | >90% |
| 3.5 | >85% |
| 3.0 | >80% |
| 2.5 | >75% |
| 2.0 | >70% |
| 1.5 | >65% |
| 1.0 | >60% |
| 0.0 | <60% |



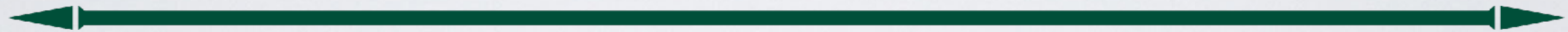
The logo is a green rectangular banner with a dark green ribbon-like shape in the center. The text "SPARTAN CODE OF HONOR" is written in white, bold, uppercase letters on the top part of the ribbon. Below it, "ACADEMIC PLEDGE" is written in smaller, white, uppercase letters. At the bottom of the ribbon, "FOR STUDENTS, BY STUDENTS" is written in a smaller, white, italicized font.

SPARTAN CODE OF HONOR

ACADEMIC PLEDGE

*FOR STUDENTS, BY STUDENTS*

# SPARTAN CODE OF HONOR



AS SPARTANS, WE CONTINUALLY STRIVE TO HOLD OURSELVES TO  
VALUES OF THE HIGHEST ETHICAL HONOR. AS STUDENTS, WE WILL  
EMPOWER AN ATMOSPHERE AS FUTURE LEADERS  
TO UPHOLD HONESTY IN OUR WORK, HOLD OUR PEERS ACCOUNTABLE  
TO THE HIGHEST STANDARD, AND TAKE PRIDE THAT INTEGRITY IS  
WORTH MORE THAN OUR GRADE. WE PLEDGE THAT THESE ETHICS  
WILL CARRY BEYOND OUR TIME AS STUDENTS  
AT MICHIGAN STATE UNIVERSITY, FOSTERING AN IMPROVED  
RECOGNITION FOR PERSONAL BEST AND INTEGRITY IN OUR SOCIETY.



Please **review the pledge** at: <https://honorcode.msu.edu/about/>

# ACCOMMODATIONS

If you have a university-documented learning difficulty or require other accommodations, please provide me with your Accommodation Letter as soon as possible and speak with me about how I can assist you in your learning.

If you do not have an Accommodation Letter but have been documented with a learning difficulty or other problems for which you may still require accommodation, please contact MSU's Resource Center for People with Disabilities (355-9642) in order to acquire current documentation.



# MICROSOFT TEAMS

- We will be using Microsoft Teams in this course.
- The primary goal of Teams is to support communication between the students, but it will also enable instructor support.
- Use the Teams channel to show ideas/tips/plots or to ask questions about syntax, error messages, etc.
- The Teams usage rules are in the syllabus.





# THE MAIN TOOL WE WILL USE: PYTHON

- In this course, we will cover most aspects of Python.
- If you want more advanced Python knowledge, additional software management skills, or opportunities to focus on applying new techniques to your research when this course is completed, please consider CMSE 802.
- The environment we will work in is called a “Jupyter Notebook”, and we will walk through one in class today.

# GETTING STARTED

- The syllabus is available on the course website:  
[cmse.msu.edu/cmse801](http://cmse.msu.edu/cmse801)
- You should have installed Anaconda on your laptops before coming to class today (ideally). You should also have received instructions for getting this set up via email. The “Software Setup Guide” is also available on the website.
  - If your local installation isn't working, you should also have access to [jupyterhub.egr.msu.edu](http://jupyterhub.egr.msu.edu) via the Engineering DECS account made available to you (again, details on the website)



# PLAN FOR TODAY

1. Getting, using, and submitting a Jupyter notebook.
  - a. go to course website folder
  - b. download the Jupyter notebook
  - c. Read the notebook, answer everything in it, and submit the final notebook into the corresponding D2L dropbox.
2. In-class activity: "Algorithms". Instructions are in the notebook.